

- This exam has 3 questions for a total of 25 marks.
- Warning: CMI's academic policy regarding cheating applies to this exam.
- You can use results seen in class without reproving them. You may use basic data structures like arrays, heaps, binary search trees, linked lists. However, if you use any other data structure, you have to describe its structure as well as justify its update and query time.
- Keep your answers precise. A short, clear, precise answer is always preferred over a long, ambiguous answer.
- Whenever you are asked to design an algorithm, you must make it as efficient as possible, also prove its correctness and analyze the running time. The credit will depend on the running time of your algorithm as well, apart from its correctness.

1. Compare each of the following pairs of functions and state whether they are related by O , o , Ω , ω , or Θ relation. Justify your answer in one or two lines. [12]

1. $\sqrt{n}, (\log n)^2$

2. $n^{\frac{3}{2}}, n^{\frac{1}{2}}$

3. $2^{\sqrt{n}}, n^2$

4. $2^n, 4^n$

5. $2^n, n!$

6. $2^n, n^{\log n}$

2. Solve the following recurrences by the method indicated: [6]

1. $T(n) = 4T(\frac{n}{2}) + n$ by the master method

2. $T(n) = T(\frac{n}{2}) + \Theta(n^2)$ by substitution method

3. $T(n) = 2T(\frac{n}{2}) + n \log n$ by recursion tree.

3. You are given a sorted array of n elements which has been *circularly shifted*. For example, $\{35, 42, 5, 12, 23, 26\}$ is a sorted array that has been circularly shifted by 2 positions. [7]

Give an $O(\log n)$ time algorithm to find the largest element in a circularly shifted array. (The number of positions through which it has been shifted is unknown to you.)

→ Distinct elements